

Chapter 1

Artificial Intelligence

Area of Study 3: Computer science: past and present

Learning Intentions

- **Key knowledge**

- philosophical conceptions of artificial intelligence, including the Turing Test, weak AI and strong AI
- Searle's Chinese Room Argument, including standard responses both for and against
- the concept of training algorithms using data
- the concepts of model overfitting and underfitting
- support vector machines (SVM) as margin-maximising linear classifiers, including:
 - * the geometric interpretation of applying SVM binary classification to one- or two-dimensional data
 - * the creation of a second feature from one-dimensional data to allow linear classification
- neural networks, including:
 - * the structure of multi-layer perceptron neural networks
 - * the evaluation of outputs using forward propagation
 - * training neural networks by using iterative improvement of the edge weights to reduce the output error
 - * the factors leading to a resurgence in neural networks in the late 20th century
- ethical issues related to artificial intelligence and data-driven algorithms, including transparency, accountability, bias and machine ethics

- **Key skills**

- describe and compare the Turing Test, strong AI and weak AI as conceptions of artificial intelligence
- describe the Chinese Room Argument and mount an argument for or against it
- explain, at a high level, how data-driven algorithms can learn from data
- explain the optimisation objectives for training SVM and neural network binary classifiers
- explain how higher dimensional data can be created to allow for linear classification
- describe the structure of a multi-layer perceptron neural network
- evaluate the output of a small multi-layer perceptron neural network using forward propagation
- explain the consequences of model overfitting or underfitting
- explain and discuss ethical issues related to artificial intelligence and data-driven algorithms

WHAT IS ARTIFICIAL INTELLIGENCE?

John McCarthy

Computer Science Department

Stanford University

Stanford, CA 94305

`jmc@cs.stanford.edu`

`http://www-formal.stanford.edu/jmc/`

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Abstract

This article for the layman answers basic questions about artificial intelligence. The opinions expressed here are not all consensus opinion among researchers in AI.

1 Basic Questions

Q. What is artificial intelligence?

A. It is the science and engineering of making intelligent machines, especially intelligent computer programs. It is related to the similar task of using computers to understand human intelligence, but AI does not have to confine itself to methods that are biologically observable.

Q. Yes, but what is intelligence?

A. Intelligence is the computational part of the ability to achieve goals in the world. Varying kinds and degrees of intelligence occur in people, many animals and some machines.

Q. Isn't there a solid definition of intelligence that doesn't depend on relating it to human intelligence?

A. Not yet. The problem is that we cannot yet characterize in general what kinds of computational procedures we want to call intelligent. We understand some of the mechanisms of intelligence and not others.

Q. Is intelligence a single thing so that one can ask a yes or no question “Is this machine intelligent or not?”?

A. No. Intelligence involves mechanisms, and AI research has discovered how to make computers carry out some of them and not others. If doing a task requires only mechanisms that are well understood today, computer programs can give very impressive performances on these tasks. Such programs should be considered “somewhat intelligent”.

Q. Isn’t AI about simulating human intelligence?

A. Sometimes but not always or even usually. On the one hand, we can learn something about how to make machines solve problems by observing other people or just by observing our own methods. On the other hand, most work in AI involves studying the problems the world presents to intelligence rather than studying people or animals. AI researchers are free to use methods that are not observed in people or that involve much more computing than people can do.

Q. What about IQ? Do computer programs have IQs?

A. No. IQ is based on the rates at which intelligence develops in children. It is the ratio of the age at which a child normally makes a certain score to the child’s age. The scale is extended to adults in a suitable way. IQ correlates well with various measures of success or failure in life, but making computers that can score high on IQ tests would be weakly correlated with their usefulness. For example, the ability of a child to repeat back a long sequence of digits correlates well with other intellectual abilities, perhaps because it measures how much information the child can compute with at once. However, “digit span” is trivial for even extremely limited computers.

However, some of the problems on IQ tests are useful challenges for AI.

Q. What about other comparisons between human and computer intelligence?

Arthur R. Jensen [Jen98], a leading researcher in human intelligence, suggests “as a heuristic hypothesis” that all normal humans have the same intellectual mechanisms and that differences in intelligence are related to “quantitative biochemical and physiological conditions”. I see them as speed, short term memory, and the ability to form accurate and retrievable long term memories.

Whether or not Jensen is right about human intelligence, the situation in

Philosophy

Artificial Intelligence

The term “**Artificial Intelligence**” (**AI**) was first coined by John McCarthy in 1956.

- McCarthy was a computer scientist at Dartmouth College.
- He organised the **Dartmouth Summer Research Project on Artificial Intelligence** in 1956, along with Marvin Minsky, Nathaniel Rochester, and Claude Shannon.
- In the proposal, McCarthy wrote:

“The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it.”

That conference is usually considered the **birth of AI as a formal field of research**.

Searle’s Chinese Room Argument

Responses

1. The Systems Reply

- **Claim:** While the man in the room does not understand Chinese, the **whole system** (man + rulebook + symbols) does.
- **Searle’s counter:** Even if you memorised the whole system in your head, you still would not understand Chinese.

2. The Robot Reply

- **Claim:** If the program were given a body (sensors and effectors), it could ground the symbols in real-world experiences. That might count as understanding.
- **Searle’s counter:** The robot would still only manipulate symbols; adding sensors does not create semantics.

3. The Brain Simulator Reply

- **Claim:** If the program simulated the exact activity of a native Chinese speaker’s neurons, then the system would genuinely understand Chinese.
- **Searle’s counter:** Even simulating the brain is still just symbol manipulation; no actual understanding occurs.

4. The Other Minds Reply

- **Claim:** We cannot directly know if anyone else really understands language—we infer it from behaviour. So if the Chinese Room behaves like a fluent speaker, we should grant it understanding, just as we do with humans.
- **Searle’s counter:** With humans we assume biological processes create semantics, but computers are fundamentally different.

The Turing Test: The Imitation Game

Critique of the New Problem

“The new problem has the advantage of drawing a fairly sharp line between the physical and the intellectual capacities of a man.”

“The question and answer method seems to be suitable for introducing almost any one of the fields of human endeavour that we wish to include.”

Contrary Views on the Main Question

1. The Theological Objection

2. The “Heads in the Sand” Objection

“I do not think that this argument is sufficiently substantial to require refutation. Consolation would be more appropriate: perhaps this should be sought in the transmigration of souls.”

3. The Mathematical Objection

4. The Argument from Consciousness

5. Arguments from Various Disabilities

6. Lady Lovelace’s Objection

7. Argument from Continuity in the Nervous System

8. The Argument from Informality of Behaviour

9. The Argument from Extra-Sensory Perception

Learning Machines

“We may hope that machines will eventually compete with men in all purely intellectual fields. But which are the best ones to start with? Even this is a difficult decision. Many people think that a very abstract activity, like the playing of chess, would be best. It can also be maintained that it is best to provide the machine with the best sense organs that money can buy, and then teach it to understand and speak English. This process could follow the normal teaching of a child. Things would be pointed out and named, etc. Again I do not know what the right answer is, but I think both approaches should be tried.”

Why this matters (historically and now)

- The **chess path** dominated early AI (1950s–1980s): programs like Deep Blue were built to solve abstract games.
- The **child path** is closer to today’s AI (machine learning, computer vision, natural language processing). Systems like ChatGPT are direct descendants of the “teach it language like a child” idea.
- Turing basically anticipated the two great streams of AI research.

Turing’s Prediction

“I believe that in about fifty years’ time it will be possible, to programme computers, with a storage capacity of about 10^9 , to make them play the imitation game so well that an average interrogator will not have more than 70 per cent chance of making the right identification after five minutes of questioning. The original question, “Can machines think?” I believe to be too meaningless to deserve discussion. Nevertheless I believe that at the end of the century the use of words and general educated opinion will have altered so much that one will be able to speak of machines thinking without expecting to be contradicted. I believe further that no useful purpose is served by concealing these beliefs. The popular view that scientists proceed inexorably from well-established fact to well-established fact, never being influenced by any improved conjecture, is quite mistaken. Provided it is made clear which are proved facts and which are conjectures, no harm can result.”

COMPUTING MACHINERY AND INTELLIGENCE

By A. M. Turing

1. The Imitation Game

I propose to consider the question, "Can machines think?" This should begin with definitions of the meaning of the terms "machine" and "think." The definitions might be framed so as to reflect so far as possible the normal use of the words, but this attitude is dangerous. If the meaning of the words "machine" and "think" are to be found by examining how they are commonly used it is difficult to escape the conclusion that the meaning and the answer to the question, "Can machines think?" is to be sought in a statistical survey such as a Gallup poll. But this is absurd. Instead of attempting such a definition I shall replace the question by another, which is closely related to it and is expressed in relatively unambiguous words.

The new form of the problem can be described in terms of a game which we call the 'imitation game.' It is played with three people, a man (A), a woman (B), and an interrogator (C) who may be of either sex. The interrogator stays in a room apart from the other two. The object of the game for the interrogator is to determine which of the other two is the man and which is the woman. He knows them by labels X and Y, and at the end of the game he says either "X is A and Y is B" or "X is B and Y is A." The interrogator is allowed to put questions to A and B thus:

C: Will X please tell me the length of his or her hair?

Now suppose X is actually A, then A must answer. It is A's object in the game to try and cause C to make the wrong identification. His answer might therefore be:

"My hair is shingled, and the longest strands are about nine inches long."

In order that tones of voice may not help the interrogator the answers should be written, or better still, typewritten. The ideal arrangement is to have a teleprinter communicating between the two rooms. Alternatively the question and answers can be repeated by an intermediary. The object of the game for the third player (B) is to help the interrogator. The best strategy for her is probably to give truthful answers. She can add such things as "I am the woman, don't listen to him!" to her answers, but it will avail nothing as the man can make similar remarks.

We now ask the question, "What will happen when a machine takes the part of A in this game?" Will the interrogator decide wrongly as often when the game is played like this as he does when the game is played between a man and a woman? These questions replace our original, "Can machines think?"